Data Pipeline: Selection, Cleaning, Split, Features, Preparation

Internal technical documentation of the data pipeline in `src/icu_forecast/`, for the group to understand and defend the work. All numbers come from `data/processed/cleaning_report.json`, `data/processed/features_report.json`, and the training split. Only aggregate counts are shown: the MIMIC-IV data use agreement does not allow publishing patient-level rows.

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1. Task the data must serve

From the M2 proposal: for **frequently admitted inpatients**, predict an **unplanned ward-to-ICU transfer within 24 h and 48 h**, scored once a day, and flag the top 10% highest-risk patients.

This defines what the data must provide:

Need	**Why**
Admission history per patient	to define "frequently admitted"
Exact location and time of every ward and ICU stay	to build daily ward snapshots and the label
Admission type	to exclude planned (elective) ICU admissions
Clinical signals recorded **on the ward, before** the transfer	to make features that are known at prediction time

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2. Data selection

2.1 Before and after

	Before (downloaded)	**After (copied to `data/raw/`)**
Tables	31 (22 `hosp`, 9 `icu`)	11 (10 `hosp`, 1 `icu`)
Compressed size	about 12 GB	about 3.4 GB

2.2 Tables used and their role

Table	Raw rows	Role
`hosp/admissions`	546,028	cohort, admission type, admission/discharge/death times
`hosp/patients`	364,627	sex, age
`hosp/transfers`	2,413,581	ward and ICU stays with times: snapshots and label
`icu/icustays`	94,458	cross-check of ICU entry times
`hosp/services`	593,071	current clinical service (medicine, surgery, ...)
`hosp/labevents` + `d_labitems`	158,374,764	main dynamic features: 21 lab tests
`hosp/prescriptions`	20,292,611	medication orders by drug class
`hosp/diagnoses_icd` + `d_icd_diagnoses`	6,364,488	comorbidities from **earlier** admissions
`hosp/omr`	7,753,027	outpatient blood pressure, weight, BMI before admission

2.3 Tables excluded and why

Table(s)	Reason
`icu/chartevents`, `inputevents`, `outputevents`, `procedureevents`, `datetimeevents`, `ingredientevents`	Every row belongs to an ICU stay, i.e. is recorded **after** the outcome. Using them would leak the answer.
`emar`, `emar_detail`, `pharmacy`	Overlap with `prescriptions`; about 2 GB more for little new information at this stage.
`poe`, `poe_detail`	Contain the "transfer to ICU" order itself: leakage risk.
`procedures_icd`, `hcpcsevents`, `drgcodes`	Coded at discharge: unknown at prediction time.
`microbiologyevents`	Useful later (culture orders signal infection); deferred to M4.
`d_hcpcs`, `d_icd_procedures`, `provider`, `caregiver`, `d_items`	Dictionaries for unused tables.

2.4 Important consequence

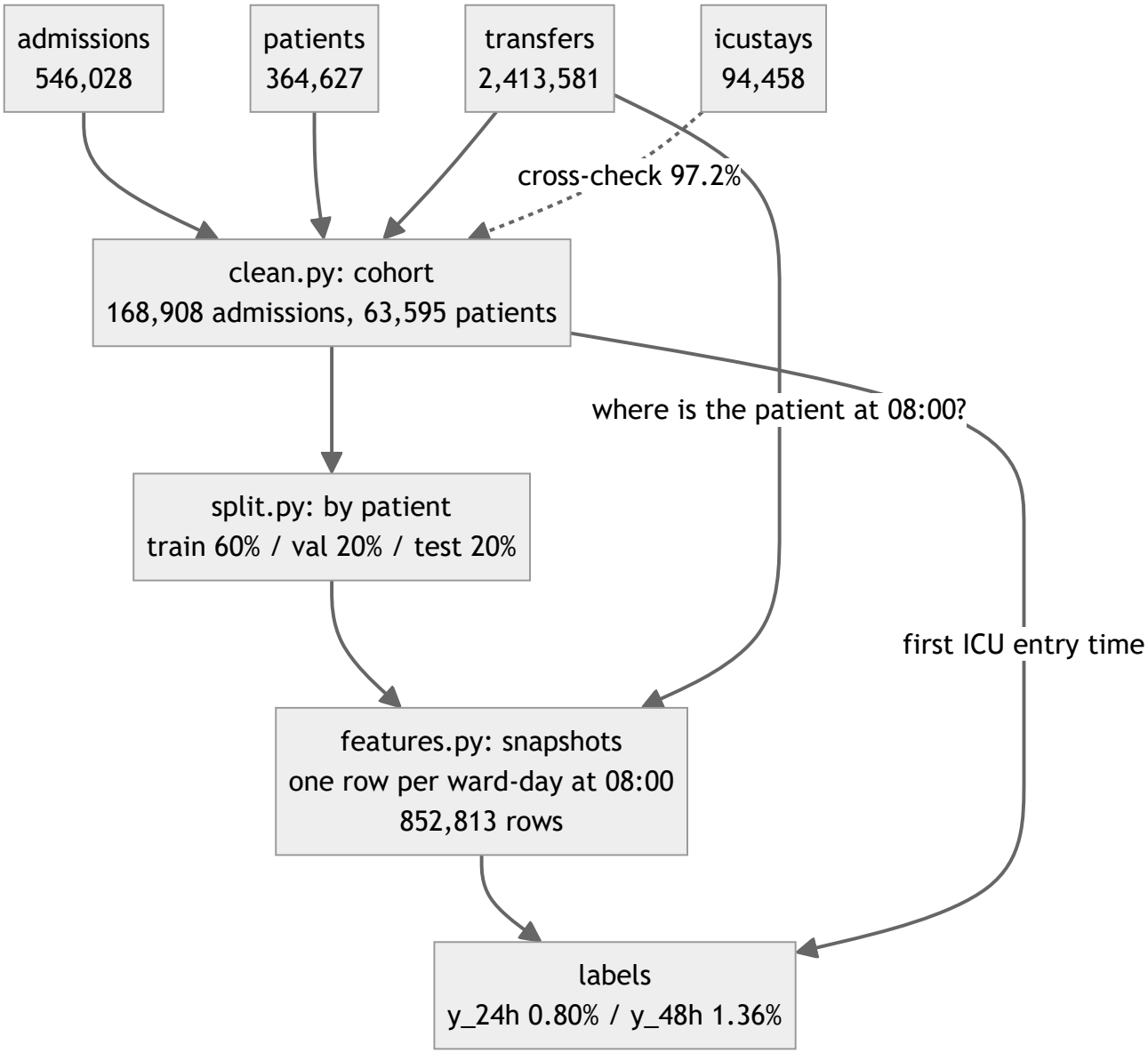
MIMIC-IV records vital signs (heart rate, blood pressure, oxygen) **only inside the ICU**. Ward patients have no vital signs in this data, so the features are driven by labs, medication orders, history, and outpatient measurements. This is a change from M2, which mentioned vital signs.

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3. Connection map: from raw tables to the LightGBM input

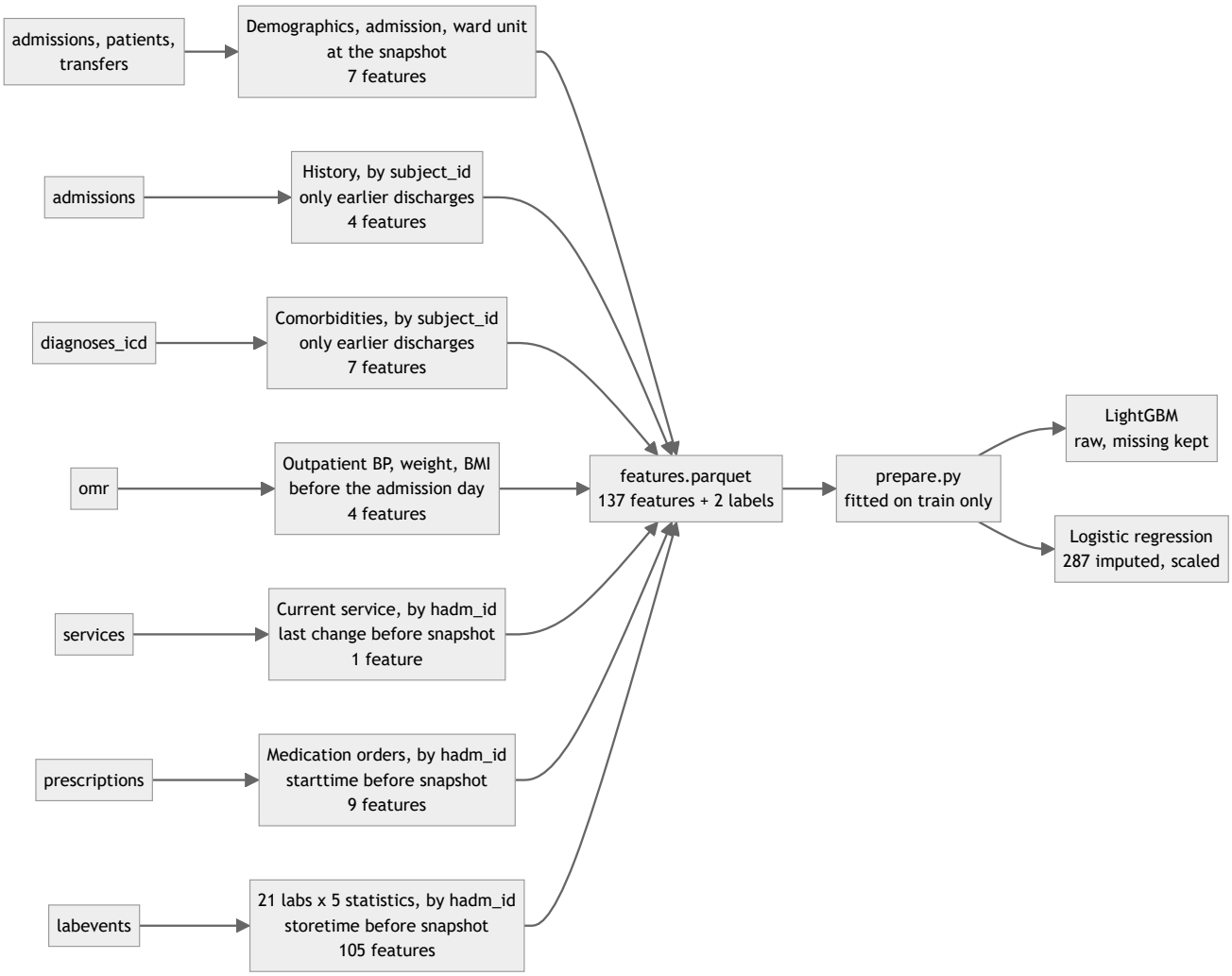
Every table is linked through two keys: `subject_id` (the patient) and `hadm_id` (one hospital admission). Time columns decide **which** rows are allowed into each snapshot.

3.1 Cohort, snapshots and labels



3.2 Feature groups feeding LightGBM

Each box on the left is a cleaned table; the middle box states the join key and the time rule that keeps the feature known at prediction time.


admissions, patients, transfers
Demographics, admission, ward unit at the snapshot 7 features
admissions
History, by subject_id only earlier discharges 4 features
diagnoses_icd
Comorbidities, by subject_id only earlier discharges 7 features
omr
Outpatient BP, weight, BMI before the admission day 4 features
services
Current service, by hadm_id last change before snapshot 1 feature
prescriptions
Medication orders, by hadm_id starttime before snapshot 9 features
labevents
21 labs x 5 statistics, by hadm_id storetime before snapshot 105 features
features.parquet 137 features + 2 labels
prepare.py fitted on train only
LightGBM raw, missing kept
Logistic regression 287 imputed, scaled

3.3 Why each feature group is an input to LightGBM

Group	Features	Source and join	Why it predicts ICU transfer	Known at prediction time because
Demographics, admission	age, sex, admission type and location, ward unit, hours since admission, days since previous discharge	`admissions`, `patients`, `transfers`	older patients and some units (cardiology, transplant) carry more risk; risk changes over the stay (Figure 6)	recorded at admission or at the snapshot
History	prior admissions (total, last 365 days), prior ICU stay, length of previous stay	`admissions` by `subject_id`	frequent and recent admissions mark fragile patients	only admissions discharged before this one
Comorbidities	heart failure, COPD, chronic kidney disease, diabetes, cancer, liver disease, sepsis	`diagnoses_icd` by `subject_id`	chronic disease raises the chance of decompensation	only diagnoses from **earlier** discharges
Outpatient	systolic and diastolic BP, weight, BMI	`omr` by `subject_id`	baseline health status	measured before the admission day
Service	current service	`services` by `hadm_id` and time	medical and surgical services differ in risk	last service change before the snapshot
Medications	new orders, new IV orders, and antibiotics, IV antibiotics, vasopressors, diuretics, opioids, insulin, anticoagulants in the last 24 h	`prescriptions` by `hadm_id`	escalation of treatment (IV antibiotics, diuretics) signals deterioration	`starttime` before the snapshot
Labs	last value, change since previous, min, max, and count over 24 h for 21 tests	`labevents` by `hadm_id`	abnormal and worsening labs (lactate, BUN, WBC) and labs being **ordered at all** signal concern (Figures 3, 4)	result **released** (`storetime`) before the snapshot

The 21 labs: creatinine, BUN, sodium, potassium, chloride, bicarbonate, anion gap, magnesium, calcium, glucose, WBC, hemoglobin, platelets, INR, bilirubin, albumin, troponin T, lactate, pH, pO2, pCO2. They cover kidney function, electrolytes and acid-base balance, infection and blood counts, liver, heart, and tissue perfusion.

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4. Configuration (`config.yaml`)

All decisions live in one file, so the code contains no hard-coded paths or numbers.

Setting	Value	Reason
`seed`	42	reproducible split and models
`split`	0.6 / 0.2 / 0.2	from M2
`readmit_window_days`	365	M2 definition of "frequently admitted"
`planned_admission_types`	ELECTIVE, SURGICAL SAME DAY ADMISSION	both are scheduled; their ICU stays are planned, not deterioration
`icu_units`	8 true ICUs (MICU, SICU, MICU/SICU, CCU, CVICU, TSICU, Neuro SICU, ICU)	step-down units (Neuro Intermediate, Neuro Stepdown) are not ICUs
`non_ward_units`	Emergency Department, ED Observation, Discharge Lounge, UNKNOWN	the patient is not on an inpatient ward there
`snapshot.hour`	8	one prediction per morning, matching bed-planning rounds
`labs`	item ID plus plausible minimum and maximum	values outside the range are recording errors
`drug_classes`	name patterns per class	groups thousands of drug names into 6 classes
`comorbidities`	ICD-9 and ICD-10 prefixes	groups diagnosis codes into 7 conditions

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5. Cleaning (`clean.py`): before and after

5.1 Admissions

Step	Before	Removed or changed	After
Raw admissions	546,028		546,028
Duplicate `hadm_id`	546,028	0	546,028
Discharge time not after admission time (impossible)	546,028	180 removed	545,848
Death time before admission time		11 set to missing	545,848
Admission overlapping the previous stay		47 flagged (not counted as readmissions)	545,848

5.2 Transfers

Step	Before	Removed	After
Raw transfers	2,413,581		2,413,581
No `hadm_id` (emergency visits without admission)	2,413,581	408,977	2,004,604
"discharge" events (not a location)	2,004,604	546,024	1,458,580
Out time before in time		0	1,458,580
Exact duplicates		3	1,458,577
Kept for cohort admissions			445,289

Check: the first ICU entry from `transfers` matches `icustays` within 1 minute for **97.2%** of admissions.

5.3 Cohort construction

Step	Rule	Admissions after	Removed
0	all valid admissions	545,848	
1	previous discharge within 365 days	236,742	309,106
2	not planned (ELECTIVE, SURGICAL SAME DAY)	214,845	21,897
3	adult (18+)	214,845	0
4	has at least one ward stay	183,190	31,655
5	not admitted straight to the ICU	**168,908**	14,282

Final cohort: 168,908 admissions, 63,595 patients, 10,377 (6.1%) with an unplanned ward-to-ICU transfer.

Why the frequent-admission rule is applied per admission (has a previous admission within 365 days) and not per patient: a per-patient rule would use **future** admissions to decide who is in the cohort, which the model cannot know at prediction time.

5.4 Labs (`labevents`, 2.6 GB)

Streamed in 64 MB blocks with pyarrow, keeping only cohort admissions and the 21 selected tests.

Step	Before	Removed or changed	After
Raw rows read	158,374,764		
Cohort admissions and 21 tests	158,374,764	145,594,359 not needed	12,780,405
No numeric value (e.g. "hemolyzed", "see comment")	12,780,405	56,089	12,724,316
Release time before draw time		39 fixed (set to draw time)	
Implausible values (outside config range)	12,724,316	342	12,723,974
Duplicate results	12,723,974	1,321	**12,722,653**

Implausible values by test: glucose 95, magnesium 69, creatinine 59, calcium 49, pO2 16, sodium 13, potassium 9, others 32. These are recording or unit errors (e.g. a pH of 70), not genuine extremes, so they are removed. Genuine extremes inside the range are kept: a lactate of 15 mmol/L is rare but real and is exactly the signal the model needs.

5.5 Prescriptions

Step	Before	Removed	After
Raw rows read	20,292,611		
Cohort admissions	20,292,611	13,371,951	6,920,660
Missing start time or drug name	6,920,660	9,754	6,910,906
Duplicate orders	6,910,906	637,782	**6,273,124**

Drug names were lower-cased and grouped into classes: anticoagulant 344,985 orders, opioid 318,794, antibiotic 283,517, insulin 266,277, diuretic 192,395, vasopressor 15,571.

5.6 Outpatient measurements (`omr`)

Step	Before	Removed	After
Rows for cohort patients	4,015,624		
Blood pressure "120/80" split into systolic and diastolic	4,015,624		more rows (one per measure)
Unparsable values		0	
Implausible (diastolic 1,148, BMI 642, weight 615, systolic 10)		2,415	
Same patient, day and measure averaged			**4,729,798**

5.7 Diagnoses and services

Table	Before	Removed	After
Diagnoses (cohort patients)		301 duplicate codes	4,304,433
Services (cohort admissions)		0 duplicates	184,113

5.8 Output files (`data/processed/`)

File	Rows	Columns
`cohort.parquet`	168,908	12
`admissions_all.parquet`	545,848	5
`transfers.parquet`	445,289	6
`labs.parquet`	12,722,653	5
`prescriptions.parquet`	6,273,124	9
`omr.parquet`	4,729,798	4
`diagnoses.parquet`	4,304,433	4
`services.parquet`	184,113	3

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6. Split (`split.py`): before and after

Before: 168,908 admissions from 63,595 patients in one table. After: every **patient** is assigned to exactly one split, stratified by whether the patient ever had a ward-to-ICU transfer, with seed 42.

Split	Patients	Admissions	Admissions with ICU transfer
Train	38,157	101,544	6.17%
Validation	12,719	33,384	6.13%
Test	12,719	33,980	6.09%

Why split by patient: one patient has several admissions and many daily snapshots. If the same patient were in both train and test, the model could memorize that patient and the test score would be too optimistic. The split is made **before** any exploration, and the test set is not used until final evaluation.

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7. Features (`features.py`): before and after

7.1 From admissions to daily snapshots

Step	Rows
Cohort admissions	168,908
Candidate 08:00 snapshots (from admission until first ICU entry, discharge or death)	864,067
Dropped: patient not on a ward at 08:00 (still in the emergency department, discharge lounge, ...)	11,254
Final snapshots	**852,813**

7.2 Illustrative example (fictional patient, not real data)

```
Day 0  14:00  admitted from the emergency department to a Medicine ward
Day 1  08:00  snapshot 1 -> features use labs released before 08:00 on day 1
Day 2  08:00  snapshot 2 -> lactate result released at 06:30 is included
Day 2  21:00  transferred to MICU
              snapshot 1: ICU transfer 61 h later -> y_24h = 0, y_48h = 0
              snapshot 2: ICU transfer 13 h later -> y_24h = 1, y_48h = 1
              no snapshot on day 3: the patient is already in the ICU
```

7.3 Labels

Label	Rule	Positive snapshots	Rate
`y_24h`	first ICU entry in (snapshot, snapshot + 24 h]	6,843	0.80%
`y_48h`	first ICU entry in (snapshot, snapshot + 48 h]	11,611	1.36%

Coverage limit: 3,524 of the 10,377 transfers (34%) happen before the first 08:00 snapshot, because a quarter of transfers occur within about 6 hours of admission. These early deteriorations cannot be predicted with morning snapshots and should be discussed in the report.

7.4 Leakage checks (all passed)

Check	Result
Snapshots at or after the ICU entry	0
Snapshots at or after discharge	0
Duplicate (admission, snapshot time) rows	0
Labs used only if released before the snapshot (`storetime`, not draw time)	by construction
Diagnoses only from earlier discharges	by construction

7.5 Output

	Before	After
Shape	8 cleaned long tables	one wide table, 852,813 rows x 142 columns
Columns		3 IDs + 2 labels + 137 features (133 numeric, 4 categorical)

8. Preparation (`prepare.py`): before and after

All statistics are fitted on the **training split only** (513,555 snapshots) and then applied to validation and test.

8.1 Two views of the same features

Model	Input	Missing values	Scaling	Categoricals
LightGBM	137 raw features	kept as missing: the model learns that "lab not ordered" is informative (Figure 4)	not needed for trees	pandas categories, levels from train
Logistic regression	287 columns	median of train + 88 missing-indicator columns	standardized (mean 0, sd 1)	one-hot: 66 columns

8.2 Before and after, training split

Feature	Before: mean	Before: sd	Before: missing	Median used to fill	After: mean	After: sd	After: missing
age (years)	60.99	18.03	0%	63	0.00	1.00	0%
hours since admission	168.1	289.6	0%	83.3	0.00	0.97	0%
creatinine, last (mg/dL)	1.41	1.58	18.6%	0.90	0.00	1.00	0%
lactate, last (mmol/L)	1.60	0.74	81.3%	1.50	0.00	1.00	0%
WBC, last (K/uL)	7.91	6.65	16.7%	7.0	0.00	0.99	0%
outpatient systolic BP (mmHg)	126.6	19.6	32.0%	125	-0.01	0.99	0%
new medication orders, 24 h	5.87	6.26	0%	4	0.00	1.00	0%

Overall, 34.8% of numeric cells in the training data are missing before preparation and 0% after.

8.3 Categorical encoding

Variable	Levels in train	One-hot columns
admission_type	7	part of 66
admission_location	11	part of 66
careunit (ward unit)	32	part of 66
curr_service	19	part of 66

Levels seen in less than 0.1% of training rows are merged into an "infrequent" column; levels never seen in training are handled without error.

8.4 Dimension reduction (`notebooks/02_pca.ipynb`)

PCA on the 13 routine labs (under 30% missing): 8 components are needed for 80% of the variance and 10 for 90%, so compression is small, and the components are harder to explain than named labs. Decision:

keep the raw lab features for LightGBM.

9. Result of the pipeline: first LightGBM test (validation split)

Model	Window	ROC-AUC	PR-AUC	Recall in top 10%	Precision in top 10%
LightGBM	24 h	0.825	0.131	48.9%	3.8%
LightGBM	48 h	0.809	0.120	45.8%	6.2%
Logistic regression	24 h	0.801	0.078	43.6%	3.4%
Logistic regression	48 h	0.785	0.079	39.5%	5.3%

The most important LightGBM inputs (Figure 8) are the ward unit, new IV medication orders, the service, time since admission, and blood-gas results, which supports the choice of transfers, prescriptions, and labevents as the core input tables.

10. How to reproduce

```
pip install -r requirements.txt
python -m icu_forecast.clean
python -m icu_forecast.split
python -m icu_forecast.features
python -m icu_forecast.prepare
python -m icu_forecast.models.lgbm
```

Appendix: figures (training split)

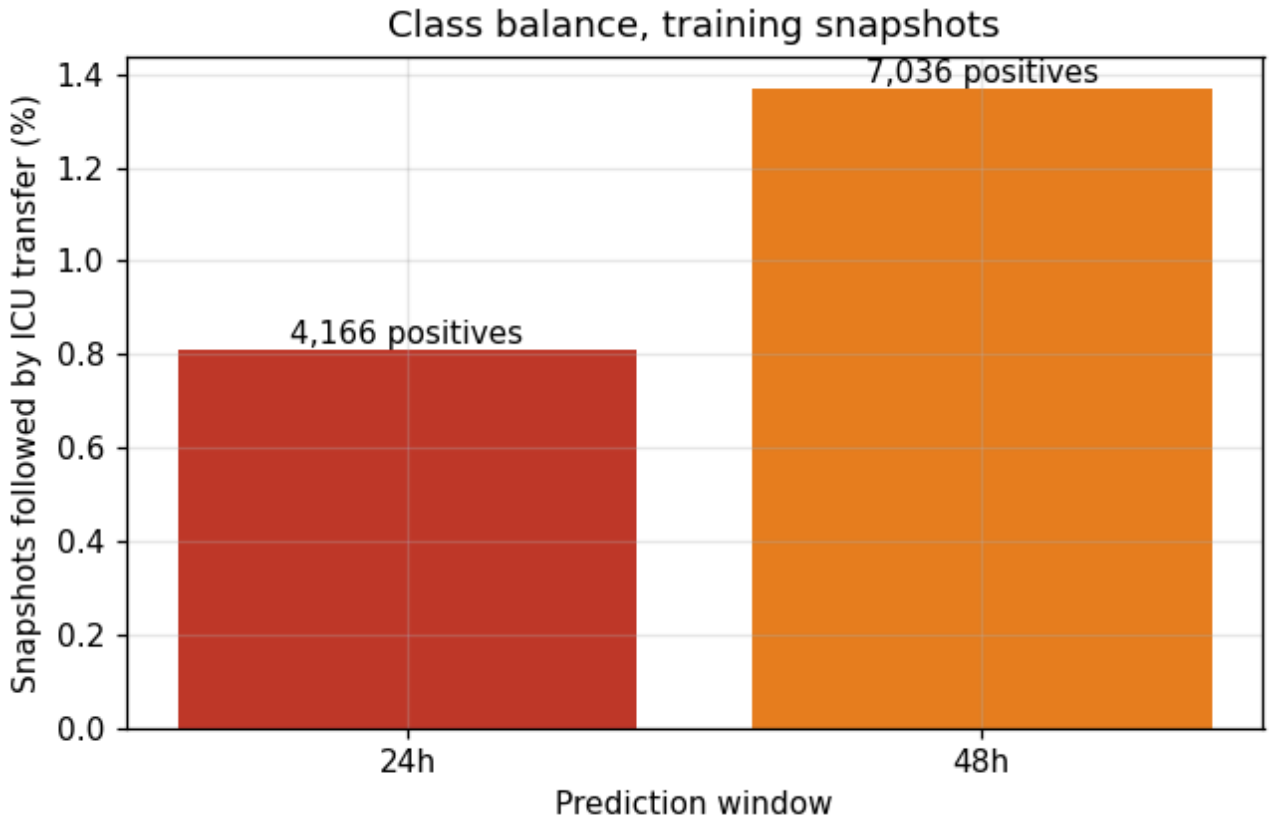


Figure 1. Share of training snapshots followed by an ICU transfer within 24 h and 48 h.

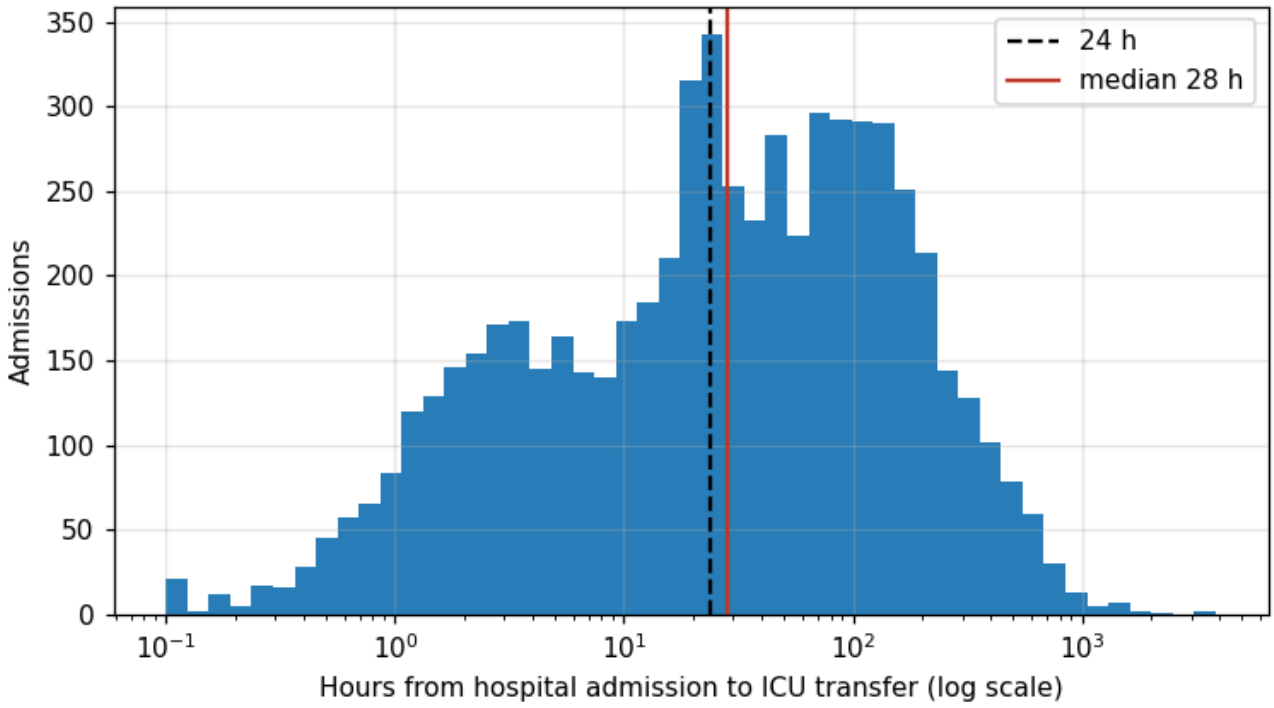


Figure 2. Hours from admission to ICU transfer; 31% of transfers happen before the first 08:00 snapshot.

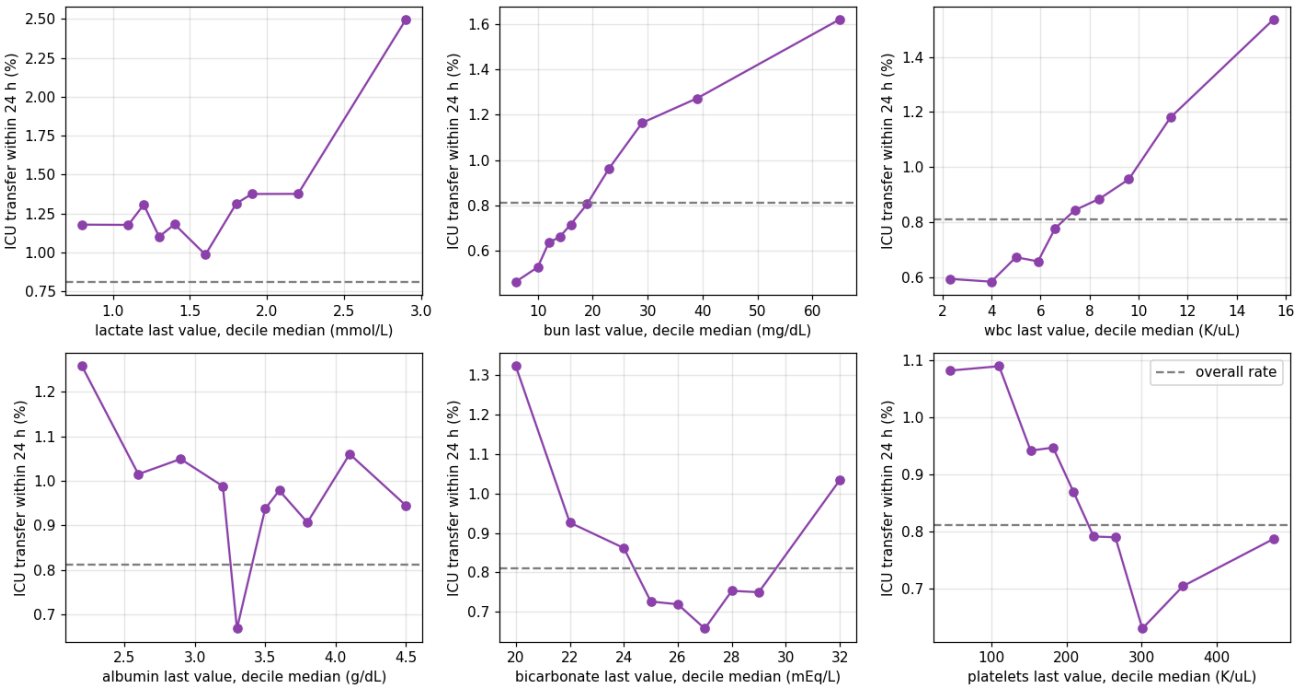


Figure 3. ICU transfer rate by decile of six key labs.

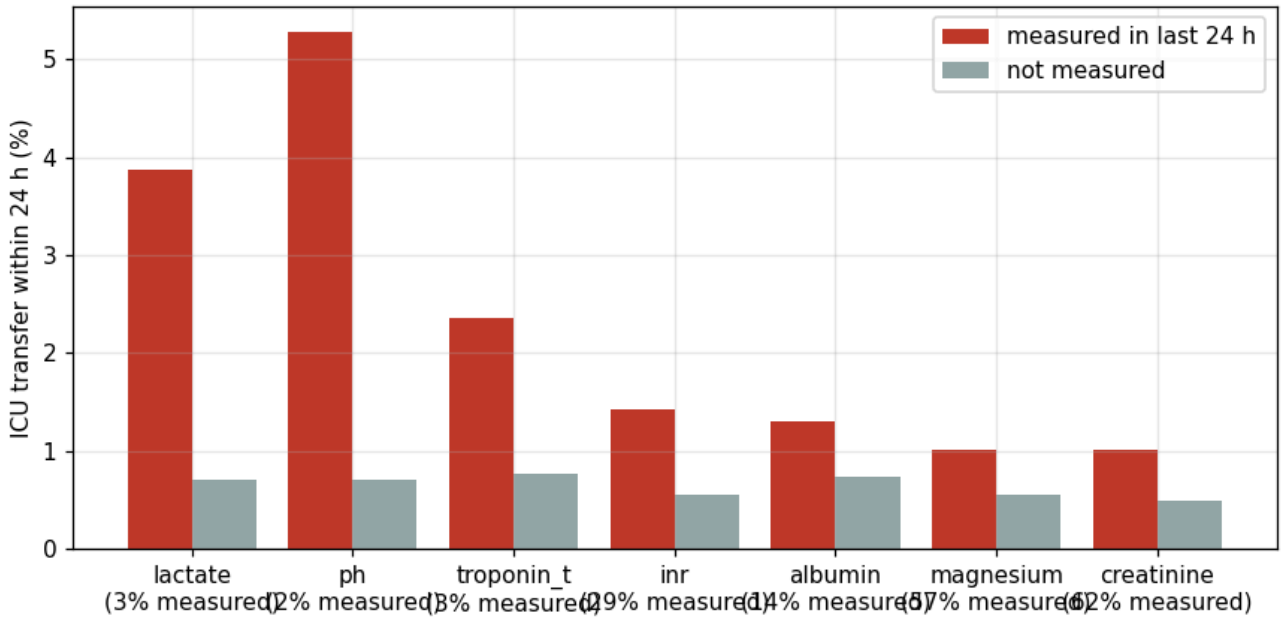


Figure 4. Transfer rate when a lab was or was not measured in the last 24 h.

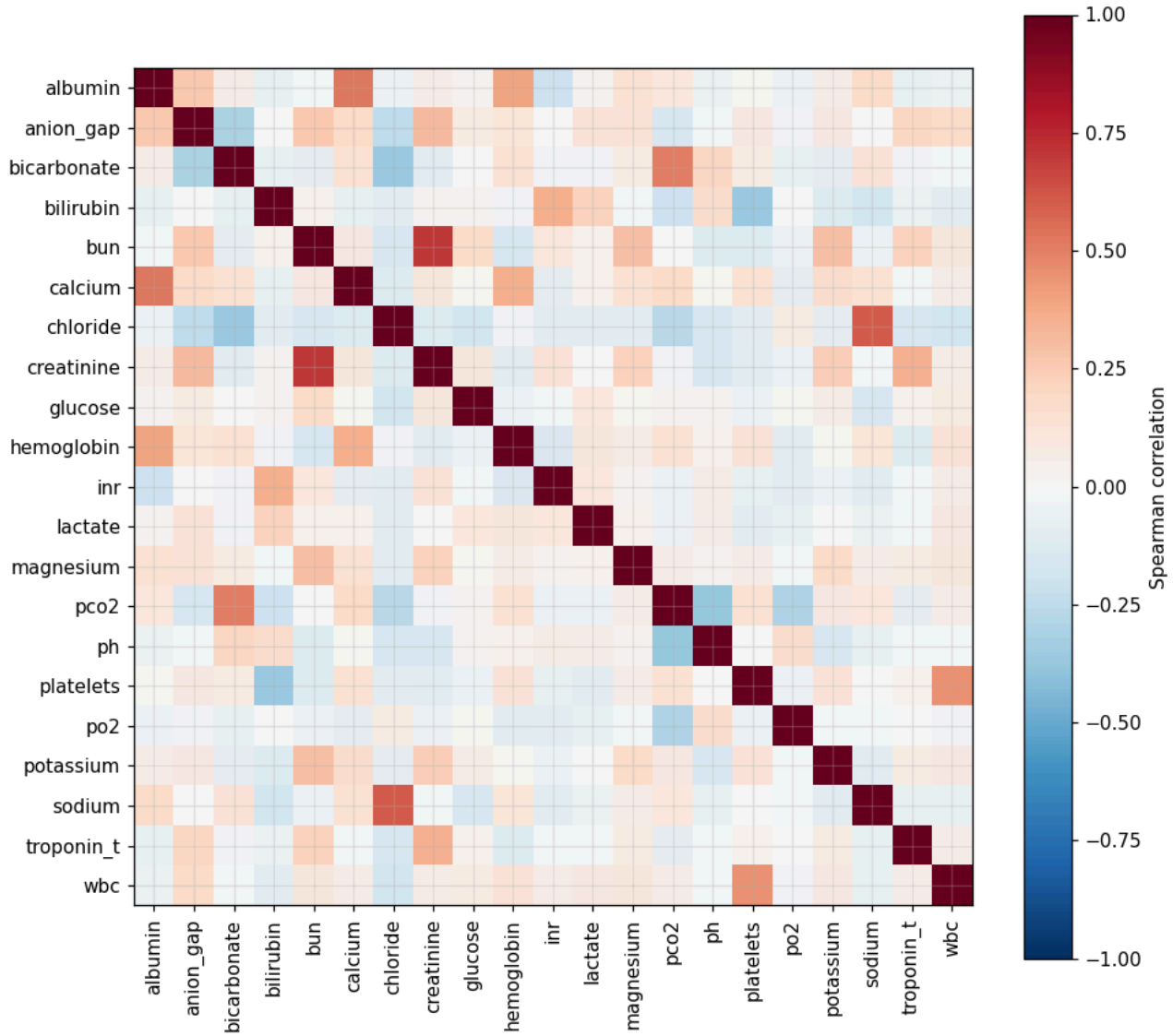


Figure 5. Spearman correlation among the latest lab values.

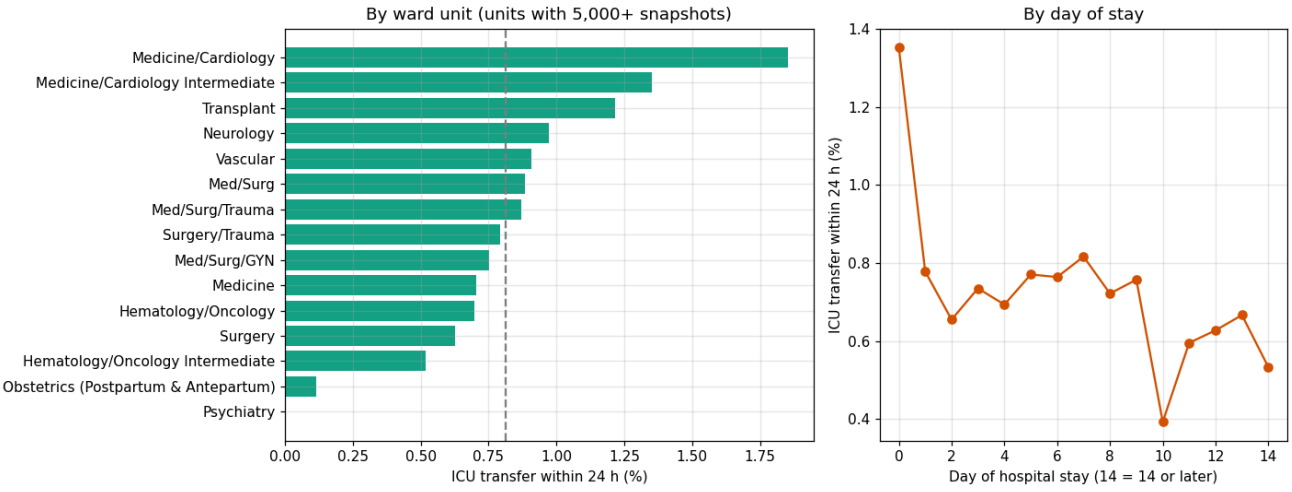


Figure 6. Transfer rate by ward unit and by day of stay.

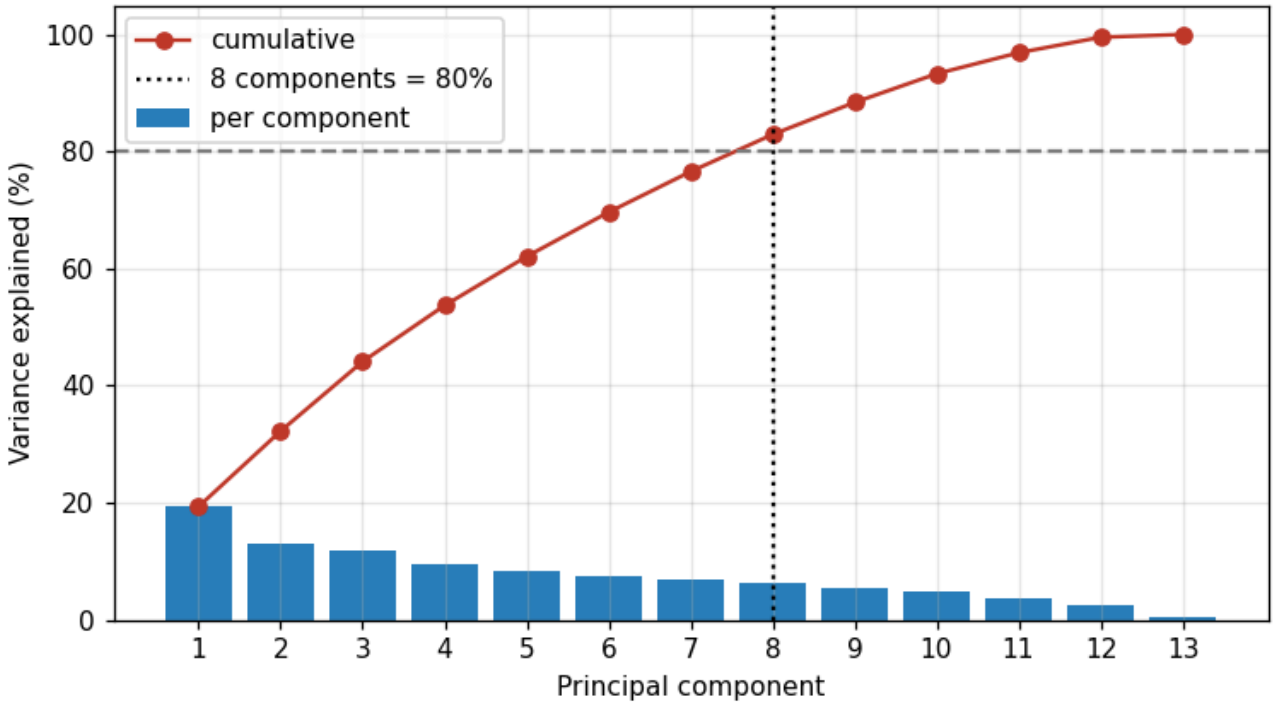


Figure 7. Variance explained by the principal components of the 13 routine labs.

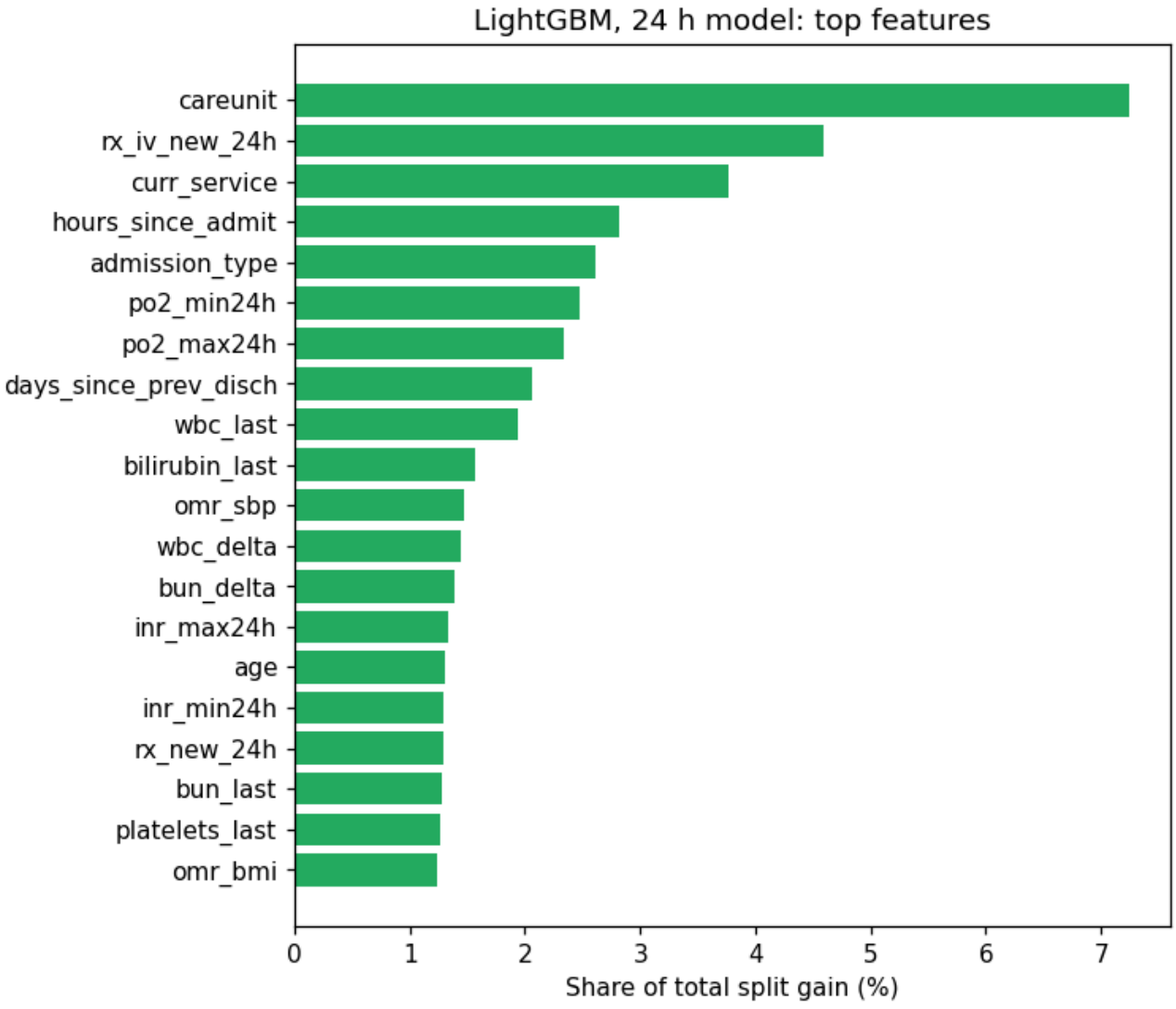


Figure 8. Top 20 LightGBM features for the 24 h model, by share of split gain.